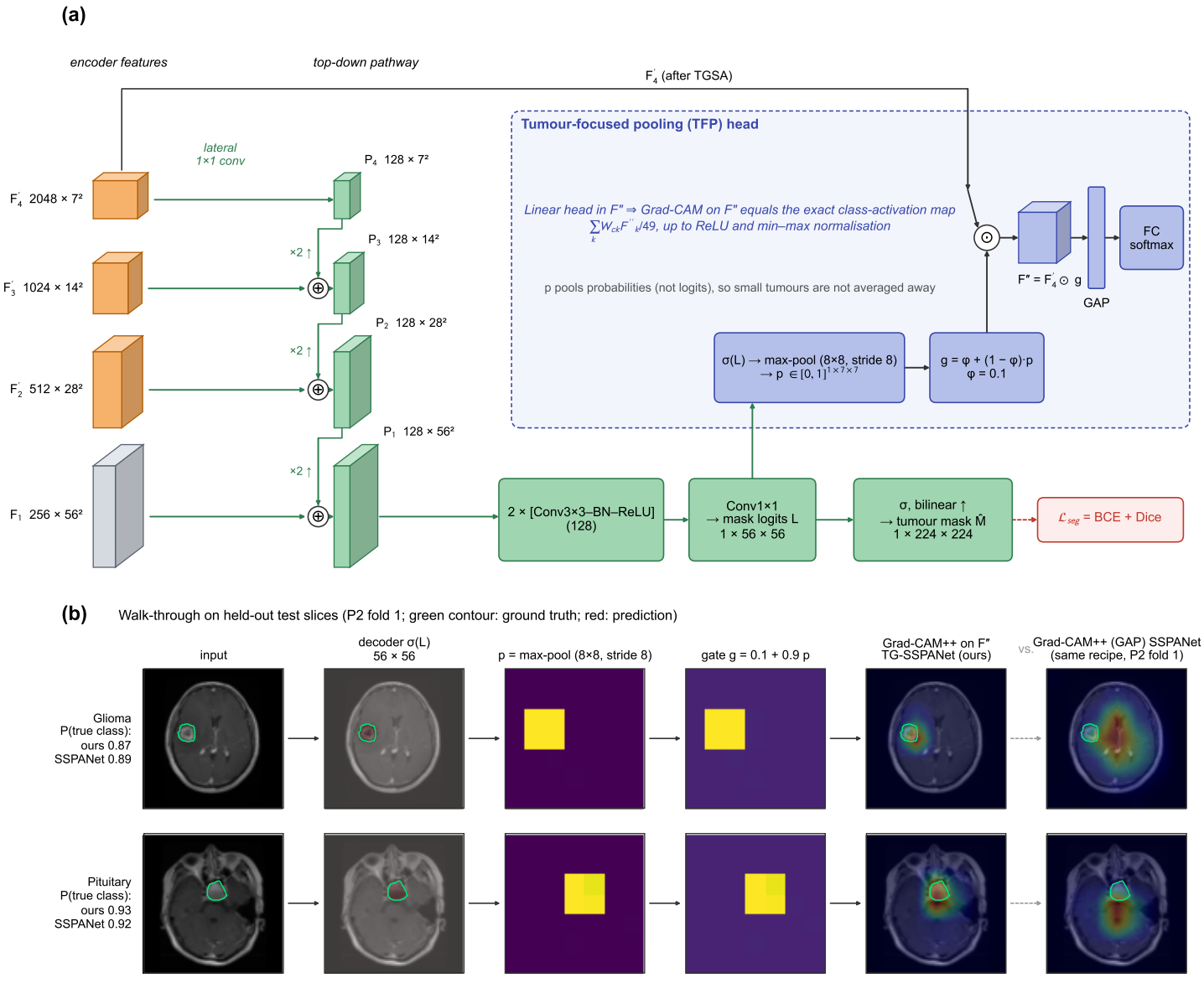

(a)
encoder features
top-down pathway
F′4 (after TGSA)
Tumour-focused pooling (TFP) head
lateral 1×1 conv
F′4 2048 × 7²
P4 128 × 7²
×2 ↑
F′3 1024 × 14²
P3 128 × 14²
×2 ↑
F′2 512 × 28²
P2 128 × 28²
×2 ↑
F1 256 × 56²
P1 128 × 56²
Linear head in F″ ⇒ Grad-CAM on F″ equals the exact class-activation map
∑k Wck F″k /49, up to ReLU and min–max normalisation
p pools probabilities (not logits), so small tumours are not averaged away
σ(L) → max-pool (8×8, stride 8) → p ∈ [0, 1]^1×7×7
g = φ + (1 − φ)·p
φ = 0.1
F″ = F′4 ⊙ g
GAP
FC softmax
2 × [Conv3×3–BN–ReLU] (128)
Conv1×1 → mask logits L 1 × 56 × 56
σ, bilinear ↑ → tumour mask M̂ 1 × 224 × 224
ℒseg = BCE + Dice
(b)
Walk-through on held-out test slices (P2 fold 1; green contour: ground truth; red: prediction)
input
decoder σ(L) 56 × 56
p = max-pool (8×8, stride 8)
gate g = 0.1 + 0.9 p
Grad-CAM++ on F″ TG-SSPANet (ours)
vs.
Grad-CAM++ (GAP) SSPANet (same recipe, P2 fold 1)
Glioma P(true class): ours 0.87 SSPANet 0.89
Pituitary P(true class): ours 0.93 SSPANet 0.92